

- 11 The density of air is 1.20 kg m^{-3} at a temperature of 20°C and standard atmospheric pressure.

Assuming that the air behaves as an ideal gas, what is the density of air at a temperature of 27°C and standard atmospheric pressure?

- A 0.89 kg m^{-3} B 1.17 kg m^{-3} C 1.23 kg m^{-3} D 1.62 kg m^{-3}

Ans: B

$$\frac{V_1}{V_2} = \frac{T_1}{T_2} \quad \text{and} \quad \frac{\rho_2}{\rho_1} = \frac{V_1}{V_2}$$

$$\therefore \frac{\rho_2}{\rho_1} = \frac{T_1}{T_2}$$

$$\frac{\rho_2}{1.20} = \frac{20 + 273.15}{27 + 273.15}$$

$$\rho_2 = 1.172 \text{ kg m}^{-3}$$

- 12 A metal block X of mass m , specific heat capacity c and temperature 80°C is placed in good